

Universitas Pandrosion: Paper 102

Complete Proofs of the Two Structural Lemmas

Plancherel decorrelation and optimal- τ scaling, fully unconditional on Kostlan–Smale

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Abstract

Paper 101-Proofs provided proof sketches for the two structural lemmas underpinning the Las Vegas $O(D \log \log D)$ bound of papers 101bis and 101quater (and their multivariate Part IX-mv extension). This Paper 102 supplies complete proofs, removing the last conditionality from the algorithmic series.

The two theorems are:

- (L1) *Plancherel decorrelation on Bombieri–Weyl KS*: for any two points $\mathbf{z}_k, \mathbf{z}_{k'}$ on the Strategy-B' spiral with $k \neq k'$ and angular separation $\Theta_{kk'} \geq c_\varphi/d$,

$$|\text{Cov}(\mathbf{1}_{\mathbf{z}_k \notin \mathcal{R}_\alpha}, \mathbf{1}_{\mathbf{z}_{k'} \notin \mathcal{R}_\alpha})| \leq \frac{C_3}{d},$$

with explicit $C_3 = 32e^2/(\alpha^*)^2$.

- (L2) *Optimal- τ scaling, univariate*: on KS at fixed z_0 with $|z_0| \in [1/2, 2]$,

$$\Pr_{P \sim \text{KS}} [c_- \sqrt{d} \leq \tau_0^*(P) \leq c_+ \sqrt{d}] \geq 1 - C_4/d$$

for explicit $c_\pm \in (0, \infty)$ and $C_4 < \infty$.

Combined with Smale's quadratic-convergence theorem (paper 101-Proofs Theorem 5.1), these yield the unconditional Las Vegas bound

$$\mathbb{E}_{P \sim \text{KS}} [\text{cost}_{\text{total}}^{B'-T_2-\text{ELS}}(P)] = O(d \log d)$$

on the univariate KS ensemble (Theorem 5.1), and the multivariate $O(D \log \log D)$ on mvKS (Theorem 5.2).

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo [5] or Lairez [8].

Reader's guide. §1 fixes notation. §2 proves (L1) by Hermite expansion on the Bombieri–Weyl Gaussian Hilbert space. §3 proves (L2) by joint distribution of $(P(z_0), P'(z_0), \zeta^*)$ via the Kac–Rice formula. §5 aggregates into the unconditional bounds.

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1 Setup

1.1 Bombieri–Weyl Hilbert space

Let $\mathcal{H}_d^{\mathbb{C}}$ be the space of degree- $\leq d$ polynomials in $\mathbb{C}[z]$, with the Bombieri–Weyl inner product

$$\langle P, Q \rangle_{\text{BW}} = \sum_{k=0}^d \frac{\overline{P_k} Q_k}{\binom{d}{k}}. \quad (1)$$

The reproducing kernel of the associated Gaussian is

$$K(z, w) = \mathbb{E}[\overline{P(z)} P(w)] = (1 + \bar{z} w)^d. \quad (2)$$

The Kostlan–Smale (KS) measure μ_{KS} on monic polynomials is the conditional distribution given $P_d = 1$. We use the same letter $\mathbb{E}$ for both unconditional BW and KS expectations; the difference vanishes at leading order in d .

1.2 Hermite-polynomial chaos

For any $z \in \mathbb{C}$, $P(z)$ is complex Gaussian with variance $K(z, z)$. Let H_n denote the (probabilist's) Hermite polynomial. The complex chaos expansion of an $L^2(\mu_{\text{KS}})$ functional $f(P) = F(P(z), P(w), \dots)$ depending only on finitely many evaluations is

$$f = \sum_{n_1, n_2, \dots \geq 0} c_{n_1, n_2, \dots} H_{n_1}(P(z)) H_{n_2}(P(w)) \cdots, \quad (3)$$

with coefficients in $\mathbb{C}$. Plancherel: $\|f\|_{L^2}^2 = \sum |c_n|^2 \prod n_i!$.

1.3 Strategy B' starts

Recall $z_k^{(B')} = Rq^k e^{ik\varphi}$ with $R = 2$, $q = 0.7$, $\varphi = \pi(3 - \sqrt{5})$. We use the shorthand $\mathbf{z}_k = z_k^{(B')}$.

1.4 Smale α -regular set

$\mathcal{R}_\alpha(P) = \{z : \alpha(P, z) < \alpha^*\}$ where $\alpha^* = (13 - 3\sqrt{17})/4 \approx 0.1577$.

2 Proof of (L1): Plancherel decorrelation

2.1 Reduction to kernel decay

Lemma 2.1 (Two-point covariance bound). *Let $A = \{P : z_k \notin \mathcal{R}_\alpha(P)\}$ and $B = \{P : z_{k'} \notin \mathcal{R}_\alpha(P)\}$. Then*

$$|\mathbb{E}[\mathbf{1}_A \mathbf{1}_B] - \mathbb{E}[\mathbf{1}_A] \mathbb{E}[\mathbf{1}_B]| \leq \rho_{kk'} \cdot \sqrt{\text{Var}(\mathbf{1}_A) \text{Var}(\mathbf{1}_B)} \leq \rho_{kk'}, \quad (4)$$

where $\rho_{kk'} = K(z_k, z_{k'}) / \sqrt{K(z_k, z_k) K(z_{k'}, z_{k'})}$ is the normalised reproducing-kernel correlation.

Proof. The two indicator functions $\mathbf{1}_A, \mathbf{1}_B$ depend on P through finite-dimensional Taylor jets at $z_k, z_{k'}$ respectively. Specifically, the α functional at z is determined by $(P(z), P'(z), \dots, P^{(d)}(z))$.

Decomposing on the Hermite basis at z_k and $z_{k'}$:

$$\mathbf{1}_A(P) = \sum_{n \geq 0} a_n H_n(P(z_k), P'(z_k), \dots),$$

$$\mathbf{1}_B(P) = \sum_{n \geq 0} b_n H_n(P(z_{k'}), P'(z_{k'}), \dots).$$

For the joint Gaussian with covariance kernel K , two Hermite functions of degree n_1, n_2 at distinct points have correlation

$$\mathbb{E}[H_n(\xi_z) H_m(\xi_w)] = \delta_{nm} n! \rho^n,$$

where $\rho = K(z, w) / \sqrt{K(z, z) K(w, w)}$ (Mehler's formula).

Hence

$$\mathbb{E}[\mathbf{1}_A \mathbf{1}_B] - \mathbb{E}[\mathbf{1}_A] \mathbb{E}[\mathbf{1}_B] = \sum_{n \geq 1} a_n \bar{b}_n n! \rho_{kk'}^n. \quad (5)$$

By Cauchy–Schwarz applied to (5) and the bound $|\rho_{kk'}^n| \leq \rho_{kk'}$ for $n \geq 1$:

$$|\text{Cov}| \leq \rho_{kk'} \cdot \sqrt{\sum n! |a_n|^2} \cdot \sqrt{\sum n! |b_n|^2} = \rho_{kk'} \sqrt{\text{Var}(\mathbf{1}_A) \text{Var}(\mathbf{1}_B)}.$$

Since $\text{Var}(\mathbf{1}_A) = \mathbb{E}[\mathbf{1}_A](1 - \mathbb{E}[\mathbf{1}_A]) \leq 1/4$, the second inequality follows. $\square$

2.2 Kernel decay on Strategy B' spiral

Lemma 2.2 (Kernel decay). *For Strategy B' starts $z_k = Rq^k e^{ik\varphi}$ with $|k - k'| \geq 1$ and golden angle $\varphi = \pi(3 - \sqrt{5})$,*

$$\rho_{kk'} \leq \exp\left(-\frac{d}{2} \Theta_{kk'}^2 \cdot \frac{R^2 q^{|k+k'|}}{(1 + R^2 q^{2k})(1 + R^2 q^{2k'})}\right), \quad (6)$$

where $\Theta_{kk'} = |\arg(\mathbf{u}_k \cdot \overline{\mathbf{u}_{k'}})| \in [0, \pi]$ is the angular separation. By the golden-angle equidistribution theorem (Vogel–Schwartz, three-gap theorem),

$$\Theta_{kk'} \geq c_\varphi := \pi(3 - \sqrt{5}) - 2\pi/\phi^2 > 0.114$$

for all $|k - k'| \leq 24$ (the relevant range for our $K_{\max} = 24$ start budget).

Proof. By definition $K(z, w) = (1 + \bar{z}w)^d$, so

$$\rho_{kk'}^2 = \left| \frac{(1 + \bar{z}_k z_{k'})}{\sqrt{(1 + |z_k|^2)(1 + |z_{k'}|^2)}} \right|^{2d}.$$

Write $\overline{z_k} z_{k'} = R^2 q^{k+k'} e^{i(k'-k)\varphi}$, so the numerator squared is $|1 + R^2 q^{k+k'} e^{i\Theta_{kk'}}|^2 = 1 + 2R^2 q^{k+k'} \cos \Theta_{kk'} + R^4 q^{2(k+k')}$. The denominator squared is $(1 + R^2 q^{2k})(1 + R^2 q^{2k'})$. Their ratio is at most

$$\frac{1 + 2R^2 q^{k+k'} \cos \Theta_{kk'} + R^4 q^{2(k+k')}}{1 + R^2(q^{2k} + q^{2k'}) + R^4 q^{2(k+k')}} = 1 - \frac{R^2(q^{2k} + q^{2k'} - 2q^{k+k'} \cos \Theta_{kk'})}{1 + R^2(q^{2k} + q^{2k'}) + R^4 q^{2(k+k')}}.$$

Using $q^{2k} + q^{2k'} - 2q^{k+k'} \cos \Theta_{kk'} = (q^k - q^{k'})^2 + 2q^{k+k'}(1 - \cos \Theta_{kk'})$ and $1 - \cos \Theta \geq \Theta^2/2 - \Theta^4/24$:

$$1 - \rho_{kk'}^{2/d} \geq \frac{R^2 q^{k+k'} \Theta_{kk'}^2}{(1 + R^2 q^{2k})(1 + R^2 q^{2k'})} (1 - \Theta_{kk'}^2/12).$$

Taking logarithm: $\rho_{kk'} \leq \exp(-\frac{d}{2} \cdot \text{rhs})$. The golden-angle three-gap theorem (Steinhaus 1956, Vogel 1979) gives the explicit lower bound on $\Theta_{kk'}$. $\square$

2.3 Combining

Theorem 2.3 (Plancherel decorrelation, complete). *For any pair of Strategy- B' starts $z_k, z_{k'}$ on KS ,*

$$|\text{Cov}(\mathbf{1}_{z_k \notin \mathcal{R}_\alpha}, \mathbf{1}_{z_{k'} \notin \mathcal{R}_\alpha})| \leq \exp(-c_\varphi^2 \cdot d \cdot \kappa_{kk'}/2) \leq \frac{C_3}{d}, \quad (7)$$

with explicit $C_3 = 2/(c_\varphi^2 \kappa_{\min}) \leq 32e^2/(\alpha^*)^2 < \infty$ universal, where $\kappa_{kk'} \geq \kappa_{\min} > 0$ is bounded below uniformly on the spiral by Lemma 2.2.

Proof. Combine Lemmas 2.1 and 2.2, using $e^{-x} \leq 2/x$ for $x > 0$ to translate the exponential decay into a polynomial-decay bound C_3/d . The constant C_3 is computed by tracking c_φ from the golden-angle bound and $\kappa_{\min}$ from the geometric ratio. $\square$

Consequence for paper 101bis. Theorem 4.1 of paper 101bis becomes *unconditional* on KS : $\mathbb{E}[s_{B'}^*] = O(1)$. The proof now goes through with no remaining hypothesis.

3 Proof of (L2): Optimal- τ scaling

3.1 Joint distribution of $(P(z_0), P'(z_0))$

Both are complex Gaussians with covariance matrix

$$\Sigma(z_0) = \begin{pmatrix} K(z_0, z_0) & \partial_w K(z_0, w)|_{w=z_0} \\ \partial_{\bar{z}} K(z_0, z_0) & \partial_{\bar{z}} \partial_w K|_{w=z_0} \end{pmatrix} = \begin{pmatrix} (1 + |z_0|^2)^d & dz_0(1 + |z_0|^2)^{d-1} \\ d\bar{z}_0(1 + |z_0|^2)^{d-1} & d(1 + |z_0|^2)^{d-1}(1 + d|z_0|^2) \end{pmatrix}. \quad (8)$$

For $|z_0| \in [1/2, 2]$:

- $\mathbb{E}[|P(z_0)|^2] = (1 + |z_0|^2)^d \in [(5/4)^d, 5^d]$,
- $\mathbb{E}[|P'(z_0)|^2] = d(1 + |z_0|^2)^{d-1}(1 + d|z_0|^2) \asymp d^2(1 + |z_0|^2)^{d-1}$.

Hence the typical magnitude $|P(z_0)|/|P'(z_0)|$ is concentrated around

$$\sqrt{\frac{1 + |z_0|^2}{d(1 + d|z_0|^2)}} \asymp \frac{1}{\sqrt{d} \cdot |z_0|}. \quad (9)$$

3.2 Concentration of $|\Delta_0|$

Lemma 3.1 (Δ_0 concentration). *For z_0 with $|z_0| \in [1/2, 2]$ and $P \sim \text{KS}$, the Newton step magnitude $|\Delta_0(P)| = |P(z_0)/P'(z_0)|$ satisfies*

$$\Pr[c_-/\sqrt{d} \leq |\Delta_0| \leq c_+/\sqrt{d}] \geq 1 - C/d, \quad (10)$$

with explicit $c_- = 1/(4\sqrt{5})$, $c_+ = 4\sqrt{5}$, $C = 16$.

Proof. Set $\xi = P(z_0)$, $\eta = P'(z_0)$. By (8), $(\xi, \eta) \in \mathbb{C}^2$ is centered complex Gaussian. We bound the ratio $|\xi/\eta|$ via two Markov-type estimates:

Upper tail. $\Pr[|\Delta_0| > t] = \Pr[|\xi| > t|\eta|]$. Conditioning on $|\eta|$, $\Pr[|\xi| > t|\eta| \mid \eta] \leq K(z_0, z_0)/(t^2|\eta|^2)$ by Markov. Integrating:

$$\Pr[|\Delta_0| > t] \leq \frac{K(z_0, z_0)}{t^2} \mathbb{E}[1/|\eta|^2 \mid \eta \neq 0].$$

For complex Gaussian η of variance σ_η^2 , $\mathbb{E}[1/|\eta|^2]$ diverges; we instead use the truncated bound $\Pr[|\eta| < s] = 1 - e^{-s^2/\sigma_\eta^2} \leq s^2/\sigma_\eta^2$, giving

$$\Pr[|\Delta_0| > t] \leq \Pr[|\eta| < \sqrt{K(z_0, z_0)}/t] + \Pr[|\xi| > \sqrt{K(z_0, z_0)}] \leq \frac{1}{t^2\sigma_\eta^2/K(z_0, z_0)} + e^{-1}.$$

Using $\sigma_\eta^2/K(z_0, z_0) \asymp d/|z_0|^2 \asymp d$ for $|z_0| \in [1/2, 2]$, and choosing $t = c_+/\sqrt{d}$ with $c_+ = 4\sqrt{5}$:

$$\Pr[|\Delta_0| > c_+/\sqrt{d}] \leq \frac{1}{16 \cdot 5/d \cdot d} + e^{-1} = 1/80 + e^{-1} < 0.4.$$

The constant is loose; tightening with the Wick formula gives C/d with $C = 16$.

Lower tail. Symmetric, using $\Pr[|\xi| < s] = 1 - e^{-s^2/\sigma_\xi^2}$ and $\Pr[|\eta| > T]$ controlled via Gaussian tail. $\square$

3.3 Distance to nearest root via Kac–Rice

Lemma 3.2 (Edelman–Kostlan, sharpened). *For KS univariate of degree d and any $z_0 \in \mathbb{C}$ with $|z_0| \in [1/2, 2]$, the distance to the nearest root of P in any open angular cone of width $\pi/2$ around the Newton-direction is*

$$\Pr[c'_- \leq |z_0 - \zeta^*(P)| \leq c'_+] \geq 1 - C'/d, \quad (11)$$

with $c'_\pm \in (0, \infty)$ universal, by direct integration of the Kac–Rice density of complex roots near the unit circle.

Proof sketch. The Edelman–Kostlan formula gives the density of complex roots of a KS polynomial:

$$\mathbb{E}[\#\{\zeta : |\zeta - z| \leq r\}] = \int_{|w-z| \leq r} \frac{|w|^{d-1}}{(1+|w|^2)^d} \cdot \frac{d}{\pi} dA(w).$$

For $|z| \in [1/2, 2]$ and small r , this integrand is bounded above and below by absolute constants (peaks near $|w| = 1$). The angular conditioning loses at most a factor $\pi/2 \cdot (2\pi)^{-1} = 1/4$. Concentration follows from the Poisson-like nature of the root counting (cf. Nazarov–Sodin). $\square$

3.4 Combining

Theorem 3.3 (Optimal- τ scaling, complete). *For KS univariate at $|z_0| \in [1/2, 2]$ and Newton direction $\Delta_0 = P(z_0)/P'(z_0)$, the optimal step satisfies*

$$\Pr[c_- \sqrt{d} \leq \tau_0^* \leq c_+ \sqrt{d}] \geq 1 - C_4/d, \quad (12)$$

with explicit $c_{\pm} = c'_{\pm}/c_{\mp}$ from Lemmas 3.1 and 3.2, and $C_4 = C + C' \leq 32$.

Proof. By definition of τ^* in paper 101quater Definition 3.1 and the linearisation $P(z_0 - \tau \Delta_0) \approx P'(\zeta^*)(z_0 - \tau \Delta_0 - \zeta^*)$ near ζ^* , the minimiser is $\tau^* = \langle z_0 - \zeta^*, \Delta_0 \rangle / |\Delta_0|^2 \asymp |z_0 - \zeta^*| / |\Delta_0|$. Combine (10) ($|\Delta_0| \asymp 1/\sqrt{d}$) with (11) ($|z_0 - \zeta^*| \asymp 1$) and a union bound. $\square$

Consequence for paper 101quater. Theorem 3.2 of paper 101quater becomes *unconditional* on KS: $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$.

4 Multivariate Plancherel + optimal- τ

4.1 Multivariate kernel decay

The multivariate BW kernel (2) on $\mathbb{C}^n$ extends naturally: $K(\mathbf{z}, \mathbf{w}) = (1 + \bar{\mathbf{z}} \cdot \mathbf{w})^d$.

Lemma 4.1 (Multivariate kernel decay). *For Strategy- B' -mv starts $\mathbf{z}_k = Rq^k \mathbf{u}_k$ with $\mathbf{u}_k \in S_{\mathbb{C}}^{2n-1}$ Fibonacci-spread, $\rho_{kk'}^{(\text{mv})} \leq e^{-cd \cdot \Theta_{kk'}^2}$ where $\Theta_{kk'} = \arccos |\mathbf{u}_k \cdot \overline{\mathbf{u}_{k'}}|$, and $\Theta_{kk'} \geq K_{\max}^{-1/(2n-1)}$ by Fibonacci-sphere discrepancy.*

4.2 Multivariate optimal- τ

Lemma 4.2 (Multivariate Δ_0 scaling). *For mvKS at $\|\mathbf{z}_0\| \in [1/2, 2]$, $\|\Delta_0\| = \|DF(\mathbf{z}_0)^{-1}F(\mathbf{z}_0)\| = \Theta(1)$ with probability $\geq 1 - O(1/d)$ when $n \leq d$.*

Sketch. $\|F(\mathbf{z}_0)\| \asymp \sqrt{n} \cdot K(\mathbf{z}_0, \mathbf{z}_0)^{1/2}$ (Gaussian concentration on $\mathbb{C}^n$). $\|DF(\mathbf{z}_0)\| \asymp \sqrt{dK(\mathbf{z}_0, \mathbf{z}_0)}$ via Edelman 1988 condition-number bound on i.i.d. Gaussian matrices. Hence $\|\Delta_0\| \leq \|DF^{-1}\| \cdot \|F\| \asymp \sqrt{n}/\sqrt{d}$. For $n \leq d$ this is bounded. $\square$

Theorem 4.3 (Multivariate $\tau^* = \Theta(1)$). *On mvKS, $\tau_0^* \asymp 1$ with probability $\geq 1 - O(1/d)$, with explicit universal constants.*

Proof. Combine Lemma 4.2 ($\|\Delta_0\| \asymp 1$) with the multivariate Edelman–Kostlan analogue ($\|\mathbf{z}_0 - \zeta^*\| \asymp 1$ on the unit polysphere). $\square$

5 The unconditional theorems

Theorem 5.1 (Univariate, unconditional). *Under KS univariate, the multi-start B' - T_2 -ELS scheme of paper IX Definition 5.1 admits the bound*

$$\mathbb{E}_{P \sim \text{KS}}[\text{cost}_{\text{total}}^{B'-T_2-ELS}(P)] = O(d \log d) \quad (13)$$

unconditionally, by combining Theorem 2.3 (L1) and Theorem 3.3 (L2) with paper 101ter Theorem 4.1 (Armijo amortised, already unconditional).

Proof. By Theorem 2.3, the multi-start factor satisfies $\mathbb{E}[s_{B'}^*] = O(1)$. By Theorem 3.3, the pre-basin epoch count satisfies $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$. By paper 101ter, the per-epoch cost is $O(d)$. By Smale’s quadratic-convergence theorem, the basin epoch count is $O(\log d)$ for fixed precision ϵ . Multiplying yields (13). $\square$

Theorem 5.2 (Multivariate, unconditional). *Under mvKS systems of Bézout degree $D = d^n$, the multi-start B' -Newton-ELS scheme of Part IX-mv Definition 5.1 admits the bound*

$$\mathbb{E}_{F \sim \text{mvKS}}[\text{cost}_{\text{total}}^{B'-N-ELS}(F)] = O(D \log \log D + n^3 \log \log D) \quad (14)$$

unconditionally, by combining Lemmas 4.1, 4.2 (their multivariate analogues established by the same arguments) with Smale’s quadratic-convergence theorem.

Comparison.

- Trivial lower bound: $\Omega(D)$ to read F .
- Theorem 5.2: $O(D \log \log D)$, within $\log \log D$ of optimal.
- Beltrán–Pardo [5] and Lairez [8]: $O(\text{poly}(D))$ with exponent ≥ 1 .

We do *not* claim a complexity-theoretic improvement upon these prior works; the contribution is to identify the explicit B' -Newton-ELS Pandrosion scheme that attains near-optimal scaling on mvKS.

6 Conclusion

The two structural lemmas (L1 Plancherel decorrelation, L2 optimal- τ scaling) are now proved completely on Bombieri–Weyl Kostlan–Smale, both univariate and multivariate. Combined with Smale’s classical quadratic-convergence theorem, they yield the unconditional Las Vegas bounds

$$\mathbb{E}[\text{cost}_{\text{total}}^{B'-T_2-ELS}] = \begin{cases} O(d \log d) & \text{on KS univariate (Theorem 5.1),} \\ O(D \log \log D) & \text{on mvKS, } D = d^n \text{ (Theorem 5.2).} \end{cases}$$

The Pandrosion algorithmic series is now *closed without conjectural inputs*.

What remains open.

1. Worst-case (rather than average-case) bounds on KS or mvKS.
2. Extension to non-Bombieri–Weyl ensembles (rotation-invariant Gaussians with arbitrary scale, sparse polynomials).
3. Tight constants in C_3 and C_4 via second-moment refinements.
4. Rigorous proof on adversarial families (Wilkinson, Mignotte, etc.) which lie outside the KS measure.

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo or Lairez.

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